(A) $f(e^{-1})$ is a local maximum (B) $f(e^{1})$ is a local minimum (C) $f(e^{-1})$ is a local minimum (D) $f(e^{1})$ is a local maximum (E) $f(e^{-1})$ is a point of inflexion

14. Using the method of integration by parts, evaluate $\int_2^5 logxdx$

(A) 0.123 (B) -0.11 (C) 1.11 (D) 0.01 (E) 2.13

15. Let $f: R \to R$ and $g: R \to R$ be given by $f(x) = sinx$ and $g(x) = x^2$. Find $(gof)(30)$
(A) $\frac{1}{4}$ (B) $-\frac{1}{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$ (E) 4

16. Given that $f(x) = x + 1$ and $g(x) = \sqrt{2x+1}$. Find the domain of the composition $(gof)(x)$. (A) $\left[-\frac{3}{2}, \infty\right)$ (B) R (C) $\left[-\frac{1}{2}, \infty\right)$ (D) $\left[-\frac{1}{2}, -\infty\right)$ (E) -R

17. What is the reduction formula for $\int cos^n x dx$? (A) $I_n = \frac{1}{n}cos^{n-1}xsinx + \frac{n(n-1)}{n}I_{n-3}$
(B) $I_n = \frac{1}{n}cos^{n-1}xsinx + \frac{(n-1)}{n}I_{n-2}$ (C) $I_n = cos^{n-1}xsinx + \frac{(n-1)}{n}I_{n-2}$
(D) $I_n = \frac{1}{n}cos^{n-1}xsinx + \frac{(n+1)}{n}I_{n-2}$ (E) $I_n = ncos^{n-1}xsinx + \frac{1}{n}I_{n-2}$

18. Given that $3x^2 + xy^3 = 4x^3y$. Find y' at (-1,0). (A) 12 (B) $\frac{1}{3}$ (C) $\frac{3}{2}$ (D) $\frac{1}{4}$ (E) -1

19. Given that $y = 6x^3\cos(6x-4)$. $find\ y'$. (A) $18x^2\sin(6x-4)$ (B) $-18x^2\sin(6x-4)$ (C) $36x^2\sin(6x-4)$ (D) $-36x^3\sin(6x-4) + 18x^2\cos(6x-4)$
(E) $-36x^3\cos(6x-4) + 18x^2\sin(6x-4)$

20. Given that $f(x) = x^2 - 1$ and $g(x) = 3x + 5$, find the inverse of the composition $(gof)(x)$. (A) $f^{-1}(x) = \sqrt{x-1}$ (B) $f^{-1}(x) = \sqrt{x+1}$ (C) $f^{-1}(x) = \frac{x-5}{3}$
(D) $f^{-1}(x) = \sqrt{\frac{x-3}{2}}$ (E) $f^{-1}(x) = \sqrt{\frac{x-2}{3}}$

21. Find the derivative of $e^{-4x}(x^3+1)$. (A) $-4e^{-4x}(x^3+1)$ (B) $3e^{-4x}(x^3+1)$ (C) $(3x^2 - 4x^3 - 4)e^{-4x}$ (D) $(3x^2 + 4x^3 + 4)e^{-4x}$ (E) $-12e^{-4x}$

22. Evaluate $\int \frac{dx}{x^2+3x+2}$ (A) $\log(x+1) + C$ (B) $\log(x+2) + C$ (C) $\log\frac{x+2}{x+1}$ (D) $\log\frac{x+1}{x+2} + C$
(E) $\log\frac{x+1}{x+2}$

23. Evaluate $\lim_{n\to\infty}\frac{2n^3-3n^2+n}{6n^3}$. (A) $\frac{1}{2}$ (B) 0 (C) $\frac{1}{3}$ (D) 3 (E) $\frac{1}{4}$

$\int e^{-4x} = e^{-4x}$

24. Let $f(x) = \begin{cases} x^2 & if\ x \neq 0 \\ 1 & if\ x = 0 \end{cases}$ for each real number a, determine the value of continuity of f at a. (A) $a + 1$ (B) $-b^2$ (C) a^2 (D) $a - 1$ (E) $a - 2$

25. Find the derivative of the function $f(x) = \ln[(x^2-1)(x^3+2)]$ $w.r.t\ x$. (A) $f' = \frac{x^2-1}{(x^3+2)}$
(B) $f' = \frac{x^2-1}{(x^2-1)(x^3+2)}$ (C) $f' = \frac{5x^4-3x^2+4x}{(x^2-1)(x^3+2)}$ (D) $f' = \frac{1}{(x^2-1)(x^3+2)}$ (E) $f' = \frac{1}{(x^2-1)}$

26. $\int e^x[f(x) + f'(x)]dx$ is equal to (A) $e^x + C$ (b) $e^x f'(x) + C$ (c) $e^{-x} + C$ (d) $e^x f(x) + C$ (e) $f(x) + C$

27. Determine the points of inflexion of the curve $x^5 - 5x^3$. (A) (0,0), (0,-1.22) and (0,6.38) (B) (0,0.638), (0,6.38) and (0,1.22) (C) (0,0), (-1.22,-6.38) and (1.22,6.38) (D) (0,0), (-1.22,6.38) and (1.22,-6.38) (E) (0,0), (0,1) and (0,2)

28. Evaluate $\lim_{x\to 5} \sqrt[3]{3x^2 - 4x + 9}$ (A) Does not exist (B) $\sqrt[3]{19}$ (C) $\sqrt[3]{59}$ (D) 4 (E) 3

29. Evaluate $\int e^{x+4}dx$. (A) $e^x + C$ (B) $e^{4x} + C$ (C) $e^4 + C$ (D) $e^{x+4} + C$ (E) $\frac{e^{x+4}}{4} + C$

$-4xe^{-4x}$

$-4e^{-4x}(x^3+1)\times 3$ $3x^2+xy^3-4x^3y$ $= -4$

$6x + x3y^2 - 12x^2y$

$-12e^{-4x}(x^3+1)(3x^2)$ $6(-1)+(-1)(3\times 0)^2 - 12(1)^2(0)$

37. Evaluate $\int_0^{\frac{\pi}{2}} \sin x \cos x dx$ (a) $\frac{1}{4}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) 0 (e) π

38. Evaluate $\int x^n dx$ (a) $nx^{n-1}+c$ (b) $\frac{x^{n-1}}{n}+c$ (c) $\frac{x^{n+1}}{n+1}+c$ (d) $\frac{x^{n+1}}{n}+c$ (e) $\frac{x^{n+1}}{n+1}$

39. If $y = e^x \log x$, find $\frac{dy}{dx}$ (a) $e^x(\log x)$ (b) $\frac{1}{x}e^x$ (c) $e^x\left(\frac{1}{x}+\log x\right)$ (d) none (e) $\left(\frac{1}{x}+\log x\right)$

40. Find the second order derivative of $x\cos x$ with respect to x. (a) $-x\sin x+2\cos x$ (b) $-x\cos x+2\sin x$ (c) $-(x\cos x+2\sin x)$ (d) $-x\cos x\sin x$ (e) $x\cos x-2\sin x$

41. Find the integral of 4^x with respect to x. (a) $-\frac{4^x}{\log x}+c$ (b) $\frac{4}{\log x}+c$ (c) $\frac{4^x}{\ln x}+c$ (d) $\frac{4^x}{\log x}+c$ (e) $\frac{4^x}{\log 5}+c$

42. Integrate with respect to θ, $\sqrt{1-2\sin\theta}$ (a) $\sin\theta+\cos\theta+c$ (b) $-\sin\theta+\cos\theta+c$ (c) $\sin\theta-\cos\theta+c$ (d) $-\sin\theta-\cos\theta+c$ (e) $\cos\theta+2\sin\theta+c$

43. Given $f(x)=2x^2+1$, find the range when domain $=\{-3,2,4,0\}$ (a) $\{9,19,33,1\}$ (b) $\{1,9,19,33\}$ (c) $\{19,33,9,1\}$ (d) $\{19,9,33,1\}$ (e) all of the above

44. Find the minimum and the maximum points of the curve $y=x^3-5x^2$. (a) Min. $(0,12),(1,13)$, Max. $(1,-13)$ (b) Min $(1,13),(-1,13)$, Max. $(0,12)$ (c) Min $(1,12)$, Max $(0,13),(1,-13)$ (d) Min $(0,12)$, Max $(1,13),(-1,13)$ (e) Min $(1,-13)$, Max $(0,12),(1,13)$ Min $(0,12),(1,13)$, Max $(1,-13)$

45. Differentiate $y=(\log x)^3$ with respect to x (a) $\frac{1}{x}$ (b) $3(\log x)^2$ (c) $\frac{3}{x}(\log x)^3$ (d) $\frac{3}{x}(\log x)^2$ (e) $3(\log x)^3$

46. What is the turning point for $y=x^2-3x+2$, (a) 2 (b) $\frac{3}{2}$ (c) $\frac{2}{3}$ (d) $-\frac{1}{4}$ (e) $\frac{1}{4}$

47. Evaluate $\int \sin 2x dx$. (a) $\ln|x^2|+c$ (b) $\ln|2x|+c$ (c) $\ln 2x|x^2+1|+c$ (d) $\frac{1}{2}\cos 2x+c$ (e) $-\frac{1}{2}\cos 2x+c$

48. Evaluate $\int e^{5x+7} dx$. (a) $7e^{5x+7}+c$ (b) $5e^{5x+7}+c$ (c) $e^{5x+7}+c$ (d) $\frac{1}{7}e^{5x+7}+c$ (e) $\frac{1}{5}e^{5x+7}+c$

49. Evaluate $\int \frac{2x}{x^2+1}dx$ (a) x^3+x+c (b) $\frac{x^3}{3}+x+c$ (c) $\log(x^2+1)+c$ (d) $\log|x^2|+c$ (e) $\log|x^2+1|+c$

50. If $f(x)=\sqrt{x}$, find $f'(8)$ (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{2\sqrt{2}}$ (d) $-\frac{1}{4\sqrt{2}}$ (e) $\frac{1}{4\sqrt{2}}$

51. Find dy/dx if $(x+y)^2=5$. (a) -1 (b) 1 (c) 2 (d) -2 (e) 0

52. Find the value of $\int_0^2 (4x^2-5x+7)dx$ (a) $44/3$ (b) $-44/3$ (c) $32/3$ (d) $-32/3$ (e) 22

53. Which of the following statements are correct about function:
(i) even function is symmetrical about y-axis (ii) odd function is symmetrical about x-axis
(iii) even function is symmetrical about the origin (iv) odd function is symmetrical about the origin

54. Evaluate $\int_0^{\frac{\pi}{2}} \cos x dx$. (a) 1 (b) $\frac{\pi}{2}$ (c) π (d) 2 (e) None of the above

55. Evaluate the integral $\int \cos(4x+5)dx$ (a) $\sin(4x+5)+c$ (b) $\frac{1}{4}\sin(4x+5)+C$ (c) $4\sin(4x+5)+C$ (d) $4\cos(4x+5)+C$ (e) $\frac{1}{4}\cos(4x+5)+C$

56. $\int \frac{\sec^2 x}{\tan x}dx$ is (a) $\tan x$ (b) $\ln(\tan x)$ (c) $\ln x$ (d) $\sec^2 x$ (e) $\ln(\sec x)$

DEPARTMENT OF MATHEMATICS
FEDERAL UNIVERSITY LAFIA
B.Sc. DEGREE Second Semester Examination 2018/2019 Session

TITLE: GENERAL MATHEMATICS II

INSTRUCTION: Answer all questions and use the back-side of each paper for rough work

Unit(s): 3

CODE: MTH 121 TIME: $2\frac{1}{2}$ hrs.

1. Obtain the stationary points of the function $y = x^3 + 6x^2 - 15x + 5$ (a) 1,-5 (b) 2,-3 (c) -1,5 (d) -2,3 (e) -1,-5
2. Evaluate $\lim_{x\to 0}\frac{\sin 3x}{x}$ to get (a) 3 (b) 0 (c) x (d) 1 (e) $3x$
3. The equation of the tangent to the curve $x^2 + y^2 + 3xy = 11$ at (1, 2) can be written as (a) $7y + 8x = 22$ (b) $-7y + 8x = 22$ (c) $7y - 8x = -22$ (d) $-7y - 8x = 22$ (e) $7y - 8x = 22$
4. Evaluate $\lim_{x\to 0}\frac{\sin^2 x}{1-\cos x}$ (a) 2 (b) 1 (c) -2 (d) 0 (e) ∞
5. Find the value of $\lim_{x\to 0}\frac{e^x - 1}{x}$ (a) 1 (b) e^x (c) x (d) 0 (e) Undefined
6. If A and B are two sets then a relation R from A to B is a subset of $A \times B$ if all except (a) $R = \emptyset$ (b) $R = R^{-1}$ (c) $R = A \times B$ (d) $R = A \times A$ (e) $R = \{(a, a), \forall\, a \epsilon A\}$
7. Which of the following statements is false? (a) A monotonic function on an interval is either increasing or decreasing on that interval (b) $f(x) = \frac{1}{x}$ is a constant function (c) $f(x) = \frac{x-3}{3+x}$ is a rational function (d) $f(x) = x^{\frac{1}{2}} - x^3 + x^5$ is a polynomial function (e) None of the above
8. Find y' if $y = \ln(1 + 2x)^2$ (a) $\frac{4}{(1+2x)^2}$ (b) $\frac{4}{1+2x}$ (c) $\frac{1+2x}{4}$ (d) $\frac{4}{1+2x^2}$ (e) $-\frac{4}{1+2x}$
9. Given that $y = 3\sin\theta - \sin^3\theta$, $x = \cos^3\theta$ find $\frac{dy}{dx}$ (a) $-\tan\theta$ (b) $-\cot\theta$ (c) $\tan\theta$ (d) $\cot\theta$ (e) $-\operatorname{cosec}\theta$
10. If $y = \sqrt{x^4 + 8x^2 + 1}$, then $\frac{dy}{dx} =$ (a) $\frac{4x^3+16x}{\sqrt{x^4+8x^2+1}}$ (b) $\frac{2x^3+8x}{\sqrt{x^4+8x^2+1}}$ (c) $\frac{2x^3+8x}{2\sqrt{x^4+8x^2+1}}$ (d) $\frac{1}{\sqrt{x^4+8x^2+1}}$ (e) None of the above
11. The function $f(x) = \frac{x^2-1}{x-1}$ not defined when x equals: (a) 0 (b) 1 (c) -1 (d) 2 (e) ∞
12. A function is said to be continuous at $x = a$ if $\lim_{x\to a} f(x) =$ (a) none (b) $\lim_{x\to a} f(x)$ (c) $f(a)$ (d) $f(x)$ (e) 0
13. Which of the following statements is true? (a) The function $f(x) = \sin x$ is not a continuous function (b) Polynomial functions are not continuous at all (c) The function $f(x) = \frac{x^2-4}{x+2}$ is continuous at $x = 2$ (d) The function $f(x) = \frac{x^2+2x+5}{x^2-8x+12}$ is continuous for all values of x (e) None of the above
14. Which of the following relations defined from $R \to R$ is a function? (a) $y = 3x + 2$ (b) $y = 2x^2 + 1$ (c) $y > x + 3$ (d) $y = x + 3$ (e) $y = 3x$
15. Given that $y = xe^y$, find $\frac{dy}{dx}$ (a) $\frac{1+xe^y}{e^y}$ (b) $\frac{e^y}{1+xe^y}$ (c) $\frac{e^y}{1-xe^y}$ (d) $\frac{e^{-y}}{1-xe^y}$ (e) $\frac{1-xe^y}{e^y}$
16. Evaluate $\int_2^5 \frac{1}{x}dx$ (a) $-\frac{3}{10}$ (b) $\frac{21}{100}$ (c) $-\ln 10$ (d) $\ln\frac{5}{2}$ (e) $\ln 3$

17. Find the derivative of the function $f(x)=\frac{4x+3}{2x-1}$ with respect to x, $x\neq\frac{1}{2}$.
(a) $f'(x)=-\frac{10}{2x}$ (b) $f'(x)=-\frac{10}{(2x-1)}$ (c) $f'(x)=2$ (d) $f'(x)=-\frac{10}{(2x-1)^2}$ (e) $f'(x)=\frac{10}{(2x-1)^2}$

18. The derivative of $\cos^2 x$ w.r.t. x is: (a) $-\cos x\sin x$ (b) $\cos 2x$ (c) $\sin 2x$ (d) $-\sin 2x$ (e) $2\cos x\sin x$

19. Find $\lim_{x\to 1}\frac{e^{-x}-e^{-1}}{x-1}$ (a) e^{-1} (b) e (c) $\frac{1}{e}$ (d) a and c (e) none

20. For the function $f(x)=2x+1$, find the range when the domain is (a) $\{-3,-2,-1,0,1,2,3\}$
(b) $\{-5,-3,-1,0,3,5,7\}$ (c) $\{-5,-3,-1,3,5,7\}$ (d) $\{-5,-3,-1,1,3,5,7\}$ (e) $\{3,2,0,1,2,3,4\}$

21. Find the derivative of $\frac{1}{x}e^x$ with respect to x. (a) e^x (b) $\frac{e^x}{x^2}$ (c) $\frac{x-1}{x^2}$ (d) $\frac{e^x(x-1)}{x}$ (e) $\frac{e^x(x-1)}{x^2}$

22. $\frac{1}{x\sqrt{x^2-1}}$ is the derivative of one of: (a) $\sin^{-1}x$ (b) $\cos^{-1}x$ (c) $\tan^{-1}x$ (d) $\cot^{-1}x$ (e) $\sec^{-1}x$

23. Evaluate $\lim_{x\to 0}\frac{\sin 4x}{2x}$ (a) 0 (b) undefined (c) ± 2 (d) -2 (e) 2

24. If $f(x)=\sqrt{x+1}$ and $g(x)=x^2+2$, find fog and gof.
(a) $fog=x^2, gof=x+2$ (b) $fog=x+3, gof=\sqrt{x^2+3}$ (c) $fog=\sqrt{x+1}, gof=x^2+2$
(d) $fog=x^2+2, gof=\sqrt{x+1}$ (e) $fog=\sqrt{x^2+3}, gof=x+3$

25. At what point is the function $f(x)=e^x$ continuous? (a) 1 (b) a (c) $-a$ (d) -1 (e) ∞

26. Find the area bounded by the curves $y^2=4x$ and $y=x$ where the sketch of the curves shows that the lower boundary is $y=x$ and the upper boundary is $y^2=4x$.
(a) $\frac{8}{3}$ square units (b) $-\frac{8}{3}$ square units (c) 8 square units (d) 3 square units (e) None of the above

27. Evaluate $\lim_{x\to 0}\frac{\sqrt{1+x}-\sqrt{1-x}}{x}$ (a) 1 (b) 0 (c) 2 (d) 3 (e) ∞

28. $\int\sin(ax+b)dx$ (a) $-\frac{\cos(ax+b)}{a}+C$ (b) $\frac{\cos(ax+b)}{a}+C$ (c) $-\cos(ax+b)+C$ (d) $\cos(ax+b)+C$
(e) none

29. Given that $8xy-16y^2=x^2$, find $\frac{dy}{dx}$ (a) $\frac{1}{4}$ (b) $\frac{1}{8}$ (c) $\frac{x-4y}{x+4y}$ (d) $\frac{1}{2}$ (e) $-\frac{1}{4}$

30. Evaluate $\int\sec^2 x dx$ (a) $\tan x+c$ (b) $\log\cos ecx+c$ (c) $\log|\sec x|+c$ (d) $\log|\cos ecx|+c$ (e) $\log|\cos x|+c$

31. Find the derivative of the function $y=3\sin 4x$ with respect to x.
(a) $12\sin 4x$ (b) $12\cos 4x$ (c) $12\cos x$ (d) $12\sin x$ (e) $-12\cos 4x$

32. Given $f(x)=x^2-4$, $g(x)=2x+5$, obtain $fog(x)$ (a) $4x^2+20x+25$ (b) $4x^2+20x+21$
(c) $4x^2-20x+25$ (d) $4x^2+20x-21$ (e) $4x^2-20x-21$

33. Simplify $\int\sqrt{1-\sin 2\theta}\,d\theta$. (a) $\int\sqrt{\cos\theta-\sin\theta}\,d\theta$ (b) $\pm\int(\cos\theta-\sin\theta)d\theta$
(c) $\cos\theta+c$ (d) $\pm(\sin\theta-\cos\theta)+c$ (e) $\cos\theta+\sin\theta+c$

34. Find $\int e^{x+7}dx$. (a) $7e^{x+7}+c$ (b) $e^{x+7}+c$ (c) $\frac{e^{x+7}}{8}+c$ (d) $8e^{x+7}+c$ (e) $e^{x+8}+c$

35. Find the stationary points on the curve whose parametric equations are $x=t^3$, $y=(t+1)^2$
(a) (1,0) (b) (-1,0) (c) (0,-1) (d) (0,1) (e) (-1,1)

36. $\int\sqrt[3]{x}dx$ is (a) $3\sqrt[3]{x}+K$ (b) $\frac{3}{4}\sqrt[3]{x^4}+K$ (c) $\frac{3}{4}(x\sqrt[3]{x})+K$ (d) $\frac{4}{3}(x\sqrt[3]{x})+K$ $\frac{4}{3}(\sqrt[3]{x})+K$

DEPARTMENT OF MATHEMATICS

FEDERAL UNIVERSITY, LAFIA

SECOND SEMESTER EXAMINATION 2017/2018 SESSION

Course Code: MTH 121 (Calculus) (3 Units)

Instructions: Answer All Questions Time: $2\frac{1}{2}$hrs

1. Given $f(x) = x^2 - 3x + 4$. Simplify the function $\frac{f(1+h)-f(1)}{h}$ (A) $h-1$ (B) $h(h-1)$ (C) $h-2$ (D) $h+2$ (E) h
2. Find the area enclosed between the curve $y = 2x^2 - x + 2$ and the straight line $y = 3x + 2$.
   (A) $\frac{8}{3}$ Sq.units (B) $\frac{3}{8}$ Sq.units (C) $\frac{1}{3}$ Sq.units (D) 8 Sq.units (E) $\frac{3}{17}$ Sq.units
3. Find the domain of the function $f(x) = \frac{1}{x^2-5}$ (A) $R/\{-\sqrt{5}, \sqrt{5}\}$ (B) $R/\{-5, 5\}$ (C) $R/\{-5,-5\}$ (D) $\{-5, 5\}$ (E) $\{5\}$
4. Differentiate w. r. t x, $x\sqrt{y+1} = xy^{\frac{3}{2}} + 1$. (A) $y' = \frac{(y^{3/2}-\sqrt{y+1}).\sqrt{y+1}}{(x-3x)\sqrt{y(y+1)}}$ (B) $y' = y^{3/2} - \sqrt{y+1}$ (C) $y' = \frac{\sqrt{y+1}}{(x-3x)\sqrt{y(y+1)}}$ (D) $y' = \frac{(y^{3/2}-\sqrt{y+1})}{(x-3x)\sqrt{y(y+1)}}$ (E) $y' = x\sqrt{y+1}$
5. Find the natural domain of the function $f(x) = x^2 - 5$ (A) R (B) 5 (C) $R/\{-5\}$ (D) $\sqrt{5}$ (E) R-5
6. Find the slope of the curve with equation: $x\sin y + \cos y^2 = 1$ at the point (1,0).
   (A) 1 (B) 0 (C) 2 (D) $\sin x$ (E) $\cos y$
7. If $x = a\sin\theta$, $y = b\cos\theta$, find the area under the curve between $\theta = 0$ and $\theta = \pi$
   (A) $A = \frac{ab}{2}$ (B) $A = \frac{\pi ab}{2}$ (C) $A = \frac{\pi a}{4}$ (D) $A = \frac{\pi a^2 b}{2}$ (E) $A = \frac{\pi ab^2}{2}$
8. Find the range of the function $f(x) = 3 - \frac{1}{2x-1}$ (A) $R/\left\{\frac{1}{2}\right\}$ (B) $R/\{3\}$ (C) $R/\left\{-\frac{1}{2}\right\}$ (D) R (E) $\left\{\frac{1}{2}, -\frac{1}{2}\right\}$
9. Given that $y = x^{\sin x}, find\ \frac{dy}{dx}$. (A) $\frac{dy}{dx} = \frac{1}{x}\sin x + \ln x\cos x$ (B) $\frac{dy}{dx} = x^{\sin x}\left(\frac{1}{x}\sin x + \ln x\cos x\right)$ (C) $\frac{dy}{dx} = x^{\cos x}$ (D) $\frac{dy}{dx} = \ln x\cos x$ (E) $\frac{dy}{dx} = x^{\sin x}\left(\frac{1}{x}\sin x - \ln x\cos x\right)$
10. An integral with limits is called…….integral. (A) Indefinite (B) definite(C) simple(D) triple (E) double
11. The function $y = exp_b(x)$ is an example of -------- function (A) Linear Function (B)rational function (C)polynomial function(D)piece-wise function. (E) none of the above
12. All of the below are methods of integration except integration by…..
    (A) Parts (B) substitution (C) partial fractions (D) induction (E) reduction formulae
13. Examine the curve $f(x) = x\ln x$ for minimum or maximum.

$e^{\ln x}$ $e^{-1} = \frac{1}{e}$ $x = 1/e$

$y = x\ln x$ | $y' = uv' + vu'$

$y' = x \cdot \frac{1}{x} + \ln x \cdot 1$